Spectral Radius Minimization for Signed Squares of Cycles

Anonymous

Abstract

For even $n \geq 8$, let $C_n^2 = C_n(1,2)$ be the square of the cycle, with its distance-one and distance-two edges independently assigned signs, with no periodicity assumption on the signing, and let m_n be the minimum signed adjacency spectral radius. A distinguished twisted signing attains $\rho_-(n)$, where $\rho_-(n)^2 = 4 + 2\cos(2\pi/n) + 2\cos(4\pi/n)$, at every even order. We prove that equality holds exactly for $n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}$; equivalently, strict failure occurs precisely at 32, at 40, and at every even $n \geq 48$. Structurally, a period-eight reference phase has squared spectral edge $\eta < 8$, while its six-gap phase slip is the unique least-edge abnormal positive single gap and has an isolated rank-two squared edge $c_6 \in (\eta, 8)$. Charge-compatible separated copies of this slip, together with a discrete IMS estimate, prove eventual failure, while exact finite verification shows that continuous failure begins precisely at order 48. The finite portions are reduced to explicitly finite objects and independently verified using exact arithmetic, so floating-point evidence makes no spectral decision.

Keywords: signed graph; spectral radius; cycle square; switching equivalence; phase slip; computer-assisted proof

MSC 2020: 05C50 (primary); 05C22, 68V05 (secondary)

1 Introduction

Let $n \geq 8$ be even. We write

$$G_n = C_n^2 = C_n(1,2)$$

for the graph on $\mathbb{Z}/n\mathbb{Z}$ in which two vertices are adjacent when their cyclic distance is one or two. An edge signing is a map $\sigma : E(G_n) \to \{\pm 1\}$, and A_σ is the real symmetric matrix whose i, j entry is $\sigma(ij)$ on an edge and zero otherwise. Our problem is to determine

$$m_n = \min_{\sigma : E(G_n) \to \{\pm 1\}} \rho(A_\sigma).$$

Thus the already formed cycle square is being signed: the distance-one and distance-two edges receive signs independently. It is not the graph obtained by first signing a cycle and then squaring that signed object. Switching at vertices conjugates A_σ by a diagonal $\{\pm 1\}$-matrix, so the minimization naturally takes place over switching classes.

The problem belongs to the general signing program in spectral graph theory. Signings encode the new eigenvalues of two-lifts in the work of Bilu and Linial [3], while Belardo, Cioabă, Koolen, and Wang ask explicitly which signatures minimize the adjacency spectral radius of a fixed connected graph [2]. Interlacing families give strong existence results for one side of the signed spectrum; in the bipartite setting spectral symmetry turns this into the two-sided control used to construct Ramanujan lifts [9]. The graphs C_n^2 are nonbipartite, however, so a one-sided eigenvalue estimate does not by itself control $\rho(A_\sigma)$. Here the graph is fixed at each order and the question is the exact two-sided optimum, rather than an existence bound uniform over a broad graph class.

The problem-specific starting point is Suvagiya's preprint [10, Conjecture 3]. For $C_n(1,2)$, it identifies four distinguished switching classes in which every quadrilateral is unbalanced, computes the spectra of the two twisted classes, verifies global optimality for even orders through 18, and formulates the all-even-order assertion as Conjecture 3. We determine exactly when that twisted candidate is globally minimizing, thereby resolving and disproving the conjecture. In particular, the problem, the flux coordinates, and the twisted spectral formula are inputs from that work, not novelty claims of the present paper.

The answer is unexpectedly nonmonotone. The proposed equality holds for every even order from 8 through 30, fails first at 32, returns at $34, 36, 38$, fails again at 40, and returns once more at $42, 44, 46$. Only from 48 onward does failure persist at every even order. The first counterexample therefore does not mark the beginning of continuous failure, and the two isolated failures cannot be inferred from the eventual phase-slip mechanism developed below.

To state the classification, set

$$\rho_-(n)^2 = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n}, \qquad \theta_n = \rho_-(n)^2.$$

The distinguished class has alternating triangle flux and negative Hamilton-cycle holonomy; a direct antiperiodic Fourier calculation shows that it attains $\rho_-(n)$ for every even $n \geq 8$. This gives the upper bound $m_n \leq \rho_-(n)$ at all orders. At an order where equality holds, the reverse inequality for every signing is a separate conclusion of exact exhaustion.

Theorem 1.1 (Complete classification)**.** *For every even $n \geq 8$, the distinguished twisted signing attains $\rho_-(n)$, and*

$$m_n < \rho_-(n) \quad \Longleftrightarrow \quad n = 32, \quad n = 40, \quad \textit{or} \quad n \geq 48.$$

Equivalently,

$$m_n = \rho_-(n) \quad \Longleftrightarrow \quad n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}.$$

Theorem 1.1 classifies the truth of the proposed equality. It neither classifies all minimizing signings at the valid orders nor determines the exact value of m_n at the failing orders.

Its finite component determines the irregular small-order truth table. A different part of the argument explains why failure is eventually uninterrupted: a low-spectral periodic phase supports localized elementary interfaces, and well-separated copies of those interfaces can be closed on large rings. The assertion that continuous failure begins at 48 uses both this analytic tail and an exact finite bridge; the interface theory alone gives no such sharp onset.

The structural comparison phase is period eight and is distinct from the twisted signing that attains $\rho_-(n)$. In its quadrilateral-flux word, the positive sites occur at spacing four. Replacing one spacing by a positive gap g creates a single interface; we call $q = g - 4$ its excess charge. For a cyclic collection of gaps, the global closing law is $\sum_j q_j \equiv n \pmod 8$, whereas the local translation sector carried by one charge is $\sigma_{\mathrm{sec}}(q) = q \pmod 4$. These congruences have different roles and are not interchangeable. In particular, one, two, and three copies of the gap-six interface, whose charge is two, supply the three nonzero even residue classes needed in the finite-ring constructions.

The reference squared spectral edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

For the bilateral positive single-gap operator A_g, put $H_g = A_g^2$ and $E(g) = \sup \mathrm{Spec}(H_g)$. Define c_6 to be the unique root in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}$$

of the degree-ten polynomial

$$\begin{aligned} p_6(y) = 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ - 19968y + 256. \end{aligned}$$

The exact isolation is important: all later strict comparisons are made against rational endpoints, not against a decimal approximation.

Theorem 1.2 (Reference and single-gap spectral hierarchy)**.** *For both lifts and both orientations,*

$$E(4) = \eta < c_6 = E(6), \qquad \operatorname{Spec}_{\mathrm{ess}}(H_6) = \operatorname{Spec}(H_{\mathrm{ref}}), \qquad \dim\ker(H_6 - c_6) = 2.$$

Moreover, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

The scope of Theorem 1.2 is precise: G6 is the unique minimizer among abnormal positive single gaps. It is not asserted to minimize over arbitrary multi-gap or finite-core interfaces. The strict single-gap hierarchy makes G6 the elementary low-cost defect above the period-eight bulk. Its charge allows one, two, or three separated copies to close in the required residue classes. Exact patch identification transfers the bilateral reference/G6 bounds to those finite rings, and a discrete IMS partition controls the squared spectral top between the patches. This proves failure for every even $n \geq 240$. Exact finite verification on $48 \leq n < 240$ then shows that continuous failure begins precisely at 48.

The computer-assisted part is organized as a proof by finite reduction, not as numerical evidence. Switching quotients and local interlacing reduce the relevant cases to finite exact objects. For $8 \leq n \leq 32$, complete switching-class decisions establish the lower bounds through 30 and an exact witness at 32. For $34, 36, 38, 42, 44, 46$, a parity-lifted finite state graph is proved sound, complete, and terminating, and every terminal case is closed exactly; a separate rational positive-definiteness certificate gives the witness at 40. Finally, rational positive-definiteness certificates cover every even order from 48 through 238. Independent checkers reconstruct the finite objects and repeat the integer, rational, or algebraic decisions. Exhaustive generation and local finite-state reductions have established precedents in graph theory [7, 5], while all-order classifications often combine an analytic tail with finite closure [8, 6].

Exact signed spectral-radius classifications are known in settings where the underlying graph is also allowed to vary, including unbalanced unicyclic and bicyclic graphs [1] and connected unbalanced signed graphs of fixed order [4]. Those results make it essential to keep the present novelty claim narrow. To the best of our knowledge, no previous work gives an all-order classification of the minimizing behavior over all $\{\pm 1\}$-edge signings of the fixed circulant family $C_n(1, 2)$. The closest direct source remains Suvagiya's conjecture; the complete truth set and the elementary single-gap mechanism are the two contributions supplied here. The paper is arranged as follows. Section 2 sets up switching coordinates, the twisted attaining candidate, and the period-eight reference phase. Section 3 develops gaps, charges, and the two closing laws. Sections 4 and 5 prove the G6 edge theorem and the complete single-gap hierarchy. Section 6 passes from bilateral interfaces to finite rings by patch identification and discrete IMS localization. Section 7 closes the remaining orders by exact finite arguments, and Section 8 records the resulting perspective. The appendices collect the algebraic G6 certificate and the exact finite-classification boundary.

2 Switching Coordinates and the Reference Phase

2.1 Edge signings, switching, and Hamilton gauge

Write the vertices of $G_n = C_n(1,2)$ as $\mathbb{Z}/n\mathbb{Z}$. Let a_i be the sign of the distance-one edge $\{i, i+1\}$, and let b_i be the sign of the distance-two edge $\{i, i+2\}$, with all subscripts read cyclically. Switching by $\varepsilon = (\varepsilon_i)_{i\in\mathbb{Z}/n\mathbb{Z}} \in \{\pm 1\}^n$ changes these signs to

$$a'_i = \varepsilon_i a_i \varepsilon_{i+1}, \qquad b'_i = \varepsilon_i b_i \varepsilon_{i+2}.$$

At the matrix level this is the orthogonal conjugacy $A_{\sigma'} = D_\varepsilon A_\sigma D_\varepsilon$, where $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$. Switching therefore preserves the complete spectrum, not only the spectral radius.

Lemma 2.1 (Hamilton gauge)**.** *Every signing of G_n is switching equivalent to one for which*

$$a_0 = \cdots = a_{n-2} = 1, \qquad a_{n-1} = \alpha,$$

where

$$\alpha = \prod_{i=0}^{n-1} a_i \in \{\pm 1\}.$$

The residual sign α, the Hamilton-cycle holonomy, is invariant under switching.

Proof. Set $\varepsilon_0 = 1$ and recursively choose $\varepsilon_{i+1} = \varepsilon_i a_i$ for $0 \le i \le n-2$. Then $a'_i = 1$ along the spanning path $0, 1, \ldots, n-1$. On the closing edge,

$$a'_{n-1} = \varepsilon_{n-1} a_{n-1} \varepsilon_0 = \prod_{i=0}^{n-1} a_i.$$

The product of the signs around a cycle is unchanged by switching because every vertex factor occurs twice. □

2.2 Flux words, holonomy, and the attaining candidate

Before fixing a gauge, define the triangle flux at i by

$$\tau_i = a_i a_{i+1} b_i.$$

Each switching factor again occurs twice, so τ_i is invariant. In the Hamilton gauge of Lemma 2.1, the actual distance-two signs are

$$b_i = \tau_i \quad (0 \le i \le n-3), \qquad b_{n-2} = \alpha\tau_{n-2}, \qquad b_{n-1} = \alpha\tau_{n-1}.$$

Consequently (τ, α) parametrizes the switching classes bijectively once the vertex labels and seam are fixed. It is often cleaner to absorb the closing sign into the boundary condition. More precisely, let

$$\mathcal{U}_n^\alpha = \{u : \mathbb{Z} \to \mathbb{C} : u_{j+n} = \alpha u_j\}.$$

On this n-dimensional space the signed adjacency takes the form

$$(A_\tau u)_i = u_{i-1} + u_{i+1} + \tau_{i-2} u_{i-2} + \tau_i u_{i+2}. \tag{1}$$

This twisted-boundary realization includes every edge that crosses the Hamilton-gauge seam and is unitarily identical to the original finite signed matrix. We suppress α from A_τ when the boundary space $\mathcal{U}_n^\alpha$ is displayed explicitly.

The quadrilateral-flux word is

$$Q_i = \tau_i\tau_{i+1}.$$

It satisfies $\prod_i Q_i = 1$. Conversely, a cyclic word $Q \in \{\pm1\}^n$ has a cyclic τ-lift precisely when $\prod_i Q_i = 1$; after choosing τ_0, the recurrence

$$\tau_{i+1} = Q_i\tau_i$$

determines that lift. The two choices of τ_0 give τ and $-\tau$, while α remains independent. Thus the exact cyclic data are (τ, α), or equivalently (Q, τ_0, α). The reduced pair (Q, α) represents two switching classes, a distinction that disappears only after passing to the squared spectrum or spectral radius. The lift condition $\prod_i Q_i = 1$ and the Hamilton twist α are independent. If n is even and $(Du)_i = (-1)^i u_i$, then

$$A_{-\tau} = -DA_\tau D, \qquad H_{-\tau} = DH_\tau D, \qquad H_\tau := A_\tau^2. \tag{2}$$

The unsquared spectra of the two lifts are reflected through zero; their squared spectra and spectral radii agree.

Proposition 2.2 (Attainment by the twisted candidate)**.** *For every even $n \geq 8$, the class*

$$Q_i = -1 \quad (0 \leq i < n), \qquad \alpha = -1,$$

has a representative $\tau_i = (-1)^i$ satisfying

$$\rho(A_\tau)^2 = 4\cos^2\frac{\pi}{n} + 4\cos^2\frac{2\pi}{n} = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n} = \theta_n.$$

Consequently $m_n \leq \rho_-(n)$ for every even $n \geq 8$.

Proof. Because $\alpha = -1$, the allowed Fourier parameters on $\mathcal{U}_n^{-1}$ are

$$\vartheta_k = \frac{(2k+1)\pi}{n}, \qquad 0 \leq k < n.$$

Multiplication by $(-1)^j$ carries the mode $e^{ij\vartheta}$ to $e^{ij(\vartheta+\pi)}$. Hence the span of these two modes is invariant, and in the displayed order the restriction of A_τ is

$$B(\vartheta) = 2\begin{pmatrix} \cos\vartheta & \cos 2\vartheta \\ \cos 2\vartheta & -\cos\vartheta \end{pmatrix}.$$

Thus

$$B(\vartheta)^2 = 4(\cos^2\vartheta + \cos^2 2\vartheta)I_2.$$

With $x = \cos 2\vartheta$, the quantity in parentheses is

$$F(x) = x^2 + \frac{x}{2} + \frac{1}{2}.$$

The relevant values lie in $[-1, \cos(2\pi/n)]$. Since F is convex, its maximum is at an endpoint. Now $F(-1) = 1$, whereas $n \geq 8$ gives

$$F\left(\cos\frac{2\pi}{n}\right) \geq F\left(\frac{1}{\sqrt{2}}\right) = 1 + \frac{1}{2\sqrt{2}} > 1.$$

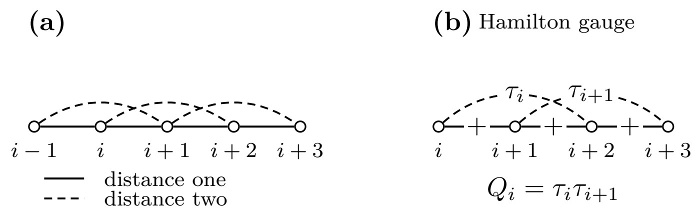

Figure 1: Switching coordinates for the cycle square. Distance-one and distance-two edges are signed independently; after Hamilton gauge, the local triangle word τ and its adjacent products $Q_i = \tau_i\tau_{i+1}$ record the flux data.

The maximum therefore occurs at $\vartheta = \pi/n$ or $\pi - \pi/n$, which yields the first equality. The double-angle identity yields the second. Since this explicit signing is in the minimization domain, the final inequality follows. □

Figure 1 summarizes the local edge types and the passage from Hamilton gauge to the adjacent flux product Q_i.

The candidate in Proposition 2.2 is the signing that attains the conjectured comparison value. It is not the period-eight reference phase introduced in Subsection 2.4. The latter is a different low-spectral bulk used to construct phase slips.

2.3 The squared local operator

For a bilateral word τ, equation (1) defines a bounded self-adjoint range-two operator on $\ell^2(\mathbb{Z})$. Its square $H_\tau = A_\tau^2$ is the nonnegative range-four operator

$$\begin{aligned}(H_\tau u)_i = 4u_i &+ u_{i-2} + u_{i+2} \\ &+ \tau_{i-4}\tau_{i-2}u_{i-4} + (\tau_{i-3} + \tau_{i-2})u_{i-3} \\ &+ (\tau_{i-2} + \tau_{i-1})u_{i-1} + (\tau_{i-1} + \tau_i)u_{i+1} \\ &+ (\tau_i + \tau_{i+1})u_{i+3} + \tau_i\tau_{i+2}u_{i+4}.\end{aligned} \tag{3}$$

Indeed, this follows by inserting equation (1) four times into $A_\tau(A_\tau u)$ and collecting equal displacements. The extreme coefficients may also be written

$$\tau_{i-4}\tau_{i-2} = Q_{i-4}Q_{i-3}, \qquad \tau_i\tau_{i+2} = Q_iQ_{i+1},$$

while, for example, $\tau_i + \tau_{i+1} = \tau_i(1 + Q_i)$. Thus a negative Q_i cancels the associated odd-displacement coupling. Formula (3) is local and precedes every periodic or interface specialization.

Translation of the coefficient word is implemented by the bilateral shift. Reflection reverses the word and is unitary after the corresponding index change. Equation (2) handles the two lifts. We will use only these explicit equivalences in the body; the fiber-level zone-folding identities are recorded with the Floquet algebra in Appendix A.

2.4 The period-eight reference phase

The canonical reference word and its quadrilateral flux are

$$\tau_{\mathrm{ref}} = (+,+,-,+,-,-,+,-), \qquad Q_{\mathrm{ref}} = (+,-,-,-,+,-,-,-),$$

extended periodically. The positive Q_{ref}-sites are exactly $4\mathbb{Z}$. More generally, for $s \in \mathbb{Z}/4\mathbb{Z}$, let B_s denote the translate whose positive Q-sites are $s + 4\mathbb{Z}$. Each B_s has two τ-lifts; since one four-site cell has Q-product -1, those lifts are antiperiodic over four sites and periodic over eight.

For a nonzero Bloch multiplier z, impose

$$u_{8m+r} = z^m v_r, \qquad 0 \le r < 8,$$

and write $A_{\mathrm{ref}}(z)$ for the resulting unsquared 8×8 fiber. It is Hermitian when $|z| = 1$.

Proposition 2.3 (Reference squared edge)**.** *Let*

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

Then

$$\sup_{|z|=1} \rho\big(A_{\mathrm{ref}}(z)\big)^2 = \eta < 8,$$

and equality occurs only at $z = 1$.

Proof. The exact determinant identity is derived and independently certified in Appendix A. We retain here the two algebraic steps that locate the edge. At $z = 1$, put $y = x^2$ and $X = y - 4$. The fiber equation reduces to

$$X^4 - 20X^2 + 80 = 0.$$

Its largest y-root is $4 + \sqrt{10 + 2\sqrt{5}} = \eta$.

For global maximality, set

$$s = \sqrt{10 + 2\sqrt{5}}, \qquad u = y - \eta, \qquad t = 2 - (z + z^{-1}).$$

On the unit circle, $t \in [0, 4]$. The same exact determinant relation, after substitution, is

$$\begin{aligned} 0 = u^4 &+ 4su^3 + 2u^2t + (40 + 12\sqrt{5})u^2 + 4sut + 8\sqrt{5}\, su \\ &+ t^2 + (4\sqrt{5} - 3)t. \end{aligned}$$

Every displayed coefficient is positive. If a squared fiber eigenvalue satisfied $y \ge \eta$, then $u, t \ge 0$, so the right-hand side could vanish only for $u = t = 0$. Thus $y \le \eta$, and equality forces $z + z^{-1} = 2$, hence $z = 1$. Finally, $\sqrt{5} < 9/4$ and $\sqrt{10 + 2\sqrt{5}} < 191/50$ give $\eta < 391/50 < 8$. □

The reference phase is also the zero-interface member of the gap language. If consecutive positive Q-sites are separated by $g = 4$, the right positive lattice is the continuation of the left lattice in the same translate B_s. Up to translation, reflection, and the lift choice, this is precisely the unperturbed reference bulk, not an abnormal interface.

3 Gaps, Charges, and Translation Sectors

3.1 Cyclic gaps and excess charge

Let $Q \in \{\pm 1\}^n$ be a cyclic quadrilateral-flux word that admits a cyclic τ-lift. Suppose first that its positive set

$$D(Q) = \{i \in \mathbb{Z}/n\mathbb{Z} : Q_i = +1\}$$

is nonempty. Choose integer representatives

$$x_1 < x_2 < \cdots < x_d < x_1 + n$$

for its d elements and put $x_{d+1} = x_1 + n$. The positive cyclic gaps and their excess charges are

$$g_j = x_{j+1} - x_j, \qquad q_j = g_j - 4 \qquad (1 \leq j \leq d).$$

The reference value is four; positive or negative q_j measures the departure of one gap from that spacing.

Proposition 3.1 (Gap and charge identities)**.** *The gap list satisfies*

$$\sum_{j=1}^{d} g_j = n, \qquad \sum_{j=1}^{d} q_j = n - 4d.$$

If n is even and Q has a cyclic τ-lift, then d is even and

$$\sum_{j=1}^{d} q_j \equiv n \pmod 8.$$

Proof. The half-open arcs from x_j to x_{j+1} partition one traversal of the n-cycle, so their lengths sum to n. Subtracting four for each of the d arcs gives the second identity.

The word has d positive and $n - d$ negative entries. Hence

$$\prod_{i=0}^{n-1} Q_i = (-1)^{n-d}.$$

A cyclic lift exists only when this product is 1. Thus $n - d$ is even; since n is even, d is even. Write $d = 2h$. The second identity becomes

$$\sum_{j=1}^{d} q_j = n - 8h,$$

which is the asserted congruence. □

The Hamilton holonomy α does not occur in this argument. It is an independent step-one cycle sign and cannot change the charge law. If $D(Q) = \varnothing$, there is no cyclic gap list and Proposition 3.1 is not invoked; this defect-free word is treated separately whenever it occurs.

3.2 Translation sectors

Recall the four reference translates B_s, $s \in \mathbb{Z}/4\mathbb{Z}$, defined by

$$(B_s)_i = +1 \iff i \equiv s \pmod 4.$$

They are sectors of the Q-word. Each has two period-eight τ-lifts, but the sector label itself depends only on the positions of the positive Q-sites.

Proposition 3.2 (Oriented gap shift)**.** *Suppose an oriented gap of length g leaves the reference sector B_s at a positive site and enters a reference sector on its right. For its charge $q = g - 4$, the right sector is*

$$B_{s+\sigma_{\text{sec}}(q)}, \qquad \sigma_{\text{sec}}(q) = q \pmod 4.$$

This law is independent of the τ-lift and Hamilton holonomy.

Proof. Let the left endpoint be $x \equiv s \pmod 4$. The next positive site is $x + g$, so the positive lattice continuing to the right is

$$x + g + 4\mathbb{Z}.$$

It is therefore B_{s+g}. Since $g = q + 4$, the residues of g and q agree modulo four. The proof uses only endpoint positions, so neither the lift sign nor α enters. □

3.3 Composition and legal cyclic closure

For consecutive oriented gaps with charges $q_1, \dots, q_k$, repeated use of Proposition 3.2 gives

$$B_s \longmapsto B_{s+q_1} \longmapsto \cdots \longmapsto B_{s+q_1+\cdots+q_k},$$

with labels reduced modulo four. Thus sector shifts add:

$$\sigma_{\text{sec}}(q_1 + \cdots + q_k) \equiv \sum_{j=1}^{k} \sigma_{\text{sec}}(q_j) \pmod 4.$$

On a cyclic gap list, $\sum_j g_j = n$, so the final reference lattice is the translate B_{s+n}; equivalently,

$$\sum_j q_j \equiv n \pmod 4.$$

This compatibility is weaker than the mod-eight law in Proposition 3.1, because it does not contain the parity information needed for a cyclic τ-lift.

The modulo-eight charge closure and the modulo-four translation-sector law encode different constraints and will be used separately. The first is a global condition on a liftable ring; the second records the local reference translate across an interface.

For later use, a gap of length six has charge $q = 6 - 4 = 2$ and sector shift two modulo four. One, two, or three such gaps contribute total charges $2, 4, 6$, respectively, and sector shifts $2, 0, 2$ modulo four. If all other gaps have the reference length four, these are precisely the charge contributions required for ring orders congruent to $2, 4, 6$ modulo eight. No spectral comparison is asserted at this stage.

4 The Elementary Six-Gap Phase Slip

4.1 The bilateral interface and its essential spectrum

The elementary interface replaces one reference gap of length four by a gap of length six. On $\mathbb{Z}$, let

$$D_6 = \{-4j : j \geq 0\} \cup \{6 + 4j : j \geq 0\}$$

and set $Q_i^{(6)} = +1$ exactly for $i \in D_6$. Choose the lift $\tau_0^{(6)} = 1$ and $\tau_{i+1}^{(6)} = Q_i^{(6)} \tau_i^{(6)}$. We write

$$(A_6 u)_i = u_{i-1} + u_{i+1} + \tau_{i-2}^{(6)} u_{i-2} + \tau_i^{(6)} u_{i+2}, \qquad H_6 = A_6^2.$$

The other lift has the same squared spectrum by equation (2); reflection gives the opposite interface orientation. The left tail has positive Q-sites $4\mathbb{Z}$, while the right tail has positive sites $6+4\mathbb{Z}$. They are translates B_0 and B_2 of the reference phase.

Every row of A_6 has four entries of absolute value one, and the matrix is real symmetric. Hence A_6 is bounded and self-adjoint, $\|A_6\| \le 4$, and

$$0 \le H_6 \le 16I.$$

Lemma 4.1 (Essential spectrum and tail matching)**.** *The G6 squared operator satisfies*

$$\operatorname{Spec}_{\mathrm{ess}}(H_6) = \operatorname{Spec}(H_{\mathrm{ref}}), \qquad \sup \operatorname{Spec}_{\mathrm{ess}}(H_6) = \eta.$$

Every $y > \eta$ in $\operatorname{Spec}(H_6)$ *is an isolated eigenvalue of finite multiplicity. Its nonzero unsquared components decay exponentially on both periodic tails and obey the stable/unstable matching criterion at $\lambda = \pm\sqrt{y}$. The next subsection writes the positive branch explicitly; the K-symmetry below identifies the negative branch.*

Proof. Choose two cuts outside the finite interface core. Decoupling at those cuts gives the direct sum of a left periodic half-line compression, a finite middle block, and a right periodic half-line compression. Since H_6 has range four, its difference from this decoupled operator has finite rank. The finite block contributes no essential spectrum.

For completeness, a periodic half-line compression has the same essential spectrum as its whole-line operator. In one direction, a whole-line Bloch solution cut off on L cells far down the half-line has norm of order $L^{1/2}$, while the residual is confined to a fixed number of boundary sites. Normalization and Weyl's criterion put every whole-line spectral point in the half-line essential spectrum. Conversely, if x lies in the whole-line resolvent and $R = (H_{\mathrm{per}} - x)^{-1}$, then the compression PRP is a two-sided inverse for the half-line operator modulo finite-rank cross-boundary terms. The half-line operator is therefore Fredholm at x. This proves the reverse inclusion.

Both whole-line tail operators are unitarily equivalent to H_{ref} by translation and diagonal switching. Finite-rank invariance now proves the displayed essential-spectrum identity, and Proposition 2.3 supplies its upper edge. Standard self-adjoint spectral theory then makes every spectral point above η discrete with finite multiplicity.

Let $H_6 u = yu$, $y > \eta$, and $\lambda = \sqrt{y}$. The orthogonal branch decomposition is

$$u_\pm = \frac{1}{2}\left(u \pm \lambda^{-1} A_6 u\right), \qquad A_6 u_\pm = \pm\lambda u_\pm.$$

At least one branch is nonzero. On either periodic tail the fourth-order recurrence has a four-dimensional eight-step monodromy. A unit-circle multiplier would give a reference Bloch state at squared energy $y > \eta$, contradicting Proposition 2.3. The reciprocal multiplier structure therefore gives two multipliers inside and two outside the unit circle. An ℓ^2 branch lies in the left unstable plane and the right stable plane; iteration gives exponential decay. The finitely many transfer maps within one period preserve the cell-state decay at individual sites. Invertibility of the recurrence proves the converse matching assertion. $\square$

4.2 Algebraic candidate and physical matching

Define

$$\begin{aligned} p_6(y) = {} & 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ & - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ & - 19968y + 256 \end{aligned}$$

and let c_6 be the unique root of p_6 in

$$J_6 = \left(\frac{7905369311620327}{10^{15}}, \frac{7905369311620328}{10^{15}}\right). \tag{4}$$

Thus $c_6 \approx 7.905369311620327$, but the rational interval, not this decimal, is used in every proof.

Fix $y > \eta$ and first take the positive branch $\lambda = \sqrt{y}$. At a left bulk cut let $U_L(\lambda)$ be the two-plane of transfer data decaying toward $-\infty$; at a right cut let $S_R(\lambda)$ be the two-plane decaying toward $+\infty$. If $D_6(\lambda)$ is transfer through the finite interface core, then

$$y \text{ is a positive-branch physical level} \quad \Longleftrightarrow \quad D_6(\lambda)U_L(\lambda) \cap S_R(\lambda) \neq \{0\}.$$

For any bases u_1, u_2 of U_L and s_1, s_2 of S_R, this is equivalent to

$$E_6(\lambda) = \det[D_6(\lambda)u_1, D_6(\lambda)u_2, s_1, s_2] = 0. \tag{5}$$

A basis change multiplies this determinant by a nonzero factor. Its zero set is therefore intrinsic; Grassmann charts are only coordinate representatives of this geometric condition.

Lemma 4.2 (Exact physical realization)**.** *The positive-branch determinant has exactly one zero over J_6; it is simple and its square is c_6. Consequently*

$$+\sqrt{c_6} \in \mathrm{Spec}(A_6), \qquad c_6 \in \mathrm{Spec}(H_6).$$

Proof. On J_6, the two stable multipliers are real, distinct, and uniformly separated from the unit circle. An exact rational interval evaluation of the genuine determinant (5) has opposite endpoint signs, and its derivative has a fixed nonzero sign throughout the interval. Hence there is exactly one simple physical positive root. At that root the matching intersection is one-dimensional: if its dimension were at least two, the 4×4 matching matrix would have rank at most two, its adjugate would vanish, and the derivative of its determinant would also vanish. Invertibility of the transfer recurrence then identifies this one-dimensional intersection with the positive eigenspace.

To identify its square, symmetric coordinates for the two stable multipliers reduce the bulk equations and the unsquared matching equation to a finite polynomial system. Exact elimination contains $p_6(y)^2$, and Sturm isolation proves that p_6 has exactly one root in J_6. Since physical existence and uniqueness were established before elimination, the squared physical root must be c_6. The interval enclosures, derivative bound, elimination identity, and Sturm count are given in Appendix A. In particular, an elimination root is not declared physical merely because it is a root of a resultant. □

4.3 The global G6 edge

It remains to prove that no other physical level has greater absolute unsquared value. The local argument above excludes another positive root through the upper endpoint of J_6. On the remaining compact energy interval up to the row-sum bound 16, the stable and unstable planes are covered by a finite exact Grassmann atlas. Exact elimination and Sturm counts give a complete finite list of algebraic candidate intervals. On every such interval the genuine unsquared determinant is bounded away from zero; a second coordinate reconstruction gives the same exclusions. The atlas also covers its chart transitions and the sole repeated multiplier, which lies away from the unit circle. These exact finite statements are proved in Appendix A; no floating-point sign is used.

Proposition 4.3 (Positive-branch completeness)**.** *The eigenvalue $+\sqrt{c_6}$ is simple, and A_6 has no positive eigenvalue larger than $\sqrt{c_6}$.*

Proof. Lemma 4.2 gives existence and simplicity. Lemma 4.1 reduces every other positive spectral point above η to the physical matching determinant. The complete exact candidate audit just described excludes all its zeros above c_6, and $\|A_6\| \le 4$ closes the squared interval above 16. $\square$

The symmetry proved in the next subsection reflects every negative unsquared level to a positive one. Combining it with Proposition 4.3 and Lemma 4.2 yields, for either lift and orientation,

$$\sup \operatorname{Spec}(H_6) = c_6. \tag{6}$$

4.4 Rank-two symmetry

The G6 word has the all-integer reflection identities

$$Q^{(6)}_{6-i} = Q^{(6)}_i, \qquad \tau^{(6)}_{7-i} = -\tau^{(6)}_i \qquad (i \in \mathbb{Z}). \tag{7}$$

The first follows directly from the two reflected positive lattices in D_6. For the second, one checks the anchor $\tau^{(6)}_7 = -\tau^{(6)}_0$. If it holds at i, then

$$\tau^{(6)}_{6-i} = Q^{(6)}_{6-i}\tau^{(6)}_{7-i} = -Q^{(6)}_i\tau^{(6)}_i = -\tau^{(6)}_{i+1}.$$

This propagates the identity in one direction, and the same recurrence backwards proves it for every integer.

Define the unitary operator

$$(Ku)_i = (-1)^i u_{9-i}.$$

Applying it twice and substituting equation (7) into the four terms of A_6 gives

$$K^2 = -I, \qquad KA_6 = -A_6K, \qquad KH_6 = H_6K.$$

Thus K maps the simple eigenspace at $+\sqrt{c_6}$ bijectively to a simple eigenspace at $-\sqrt{c_6}$, and it maps any hypothetical negative level of larger absolute value to an excluded positive level. This proves the spectral reflection used in equation (6).

Finally, self-adjointness gives the orthogonal decomposition

$$\ker(H_6 - c_6) = \ker(A_6 - \sqrt{c_6}) \oplus \ker(A_6 + \sqrt{c_6}).$$

Both summands are one-dimensional, so

$$\dim \ker(H_6 - c_6) = 2.$$

The squared level c_6 is therefore rank two; it is not simple. The two unsquared partners $+\sqrt{c_6}$ and $-\sqrt{c_6}$ are simple and their eigenvectors are exponentially localized by Lemma 4.1.

5 Optimality Among Single-Gap Interfaces

5.1 Canonical single-gap operators

For a positive integer g, let

$$D_g = \{-4j : j \ge 0\} \cup \{g + 4j : j \ge 0\}.$$

Define $Q_i^{(g)} = +1$ exactly on D_g, choose the lift $\tau_0^{(g)} = 1$, and set

$$(A_g v)_i = v_{i-1} + v_{i+1} + \tau_{i-2}^{(g)} v_{i-2} + \tau_i^{(g)} v_{i+2}, \qquad H_g = A_g^2, \qquad E(g) = \sup \operatorname{Spec}(H_g).$$

Reflection covers the reverse orientation, and equation (2) covers the other τ-lift. It is therefore enough to make each comparison for the displayed forward lift.

Gap $g = 4$ continues the same positive lattice and is the reference bulk, so $E(4) = \eta$; it is not an abnormal interface. Gap $g = 6$ is the elementary interface of Section 4, and equation (6) gives $E(6) = c_6$.

For a finitely supported vector v, extend it by zero and compute its full range-two image. Self-adjointness gives the exact Rayleigh identity

$$\frac{\langle v, H_g v\rangle}{\langle v, v\rangle} = \frac{\|A_g v\|^2}{\|v\|^2}. \tag{8}$$

The certified upper endpoint in equation (4) yields

$$T := \frac{988671163952541}{125000000000000} = \frac{7905369311620328}{10^{15}} + \frac{1}{250} > c_6 + \frac{1}{250}.$$

It is enough, therefore, to exhibit a finite vector with quotient strictly larger than T.

5.2 Exact finite-gap comparisons

The positive abnormal gaps below nine other than six are

$$g \in \{1, 2, 3, 5, 7, 8\}.$$

For each row in the following table, a displayed finite-support integer vector in the supplementary witness table has $\|v\|^2 = D$ and $\|A_g v\|^2 = N$. The final column is the exact integer

$$\Delta(N, D) = 125000000000000\, N - 988671163952541\, D.$$

g	N/D	$\Delta(N, D)$
1	812/97	5598897096603523
2	866/109	484843129173031
3	3114/393	702232566651387
5	764/96	587568260556064
7	768/97	98897096603523
8	19672/2487	174815250030533

Every entry in the last column is positive. Hence every listed quotient is strictly larger than T, and equation (8) proves

$$E(g) > c_6 + \frac{1}{250} \qquad (g \in \{1, 2, 3, 5, 7, 8\}).$$

The smallest margin is the gap-eight value

$$\frac{19672}{2487} - T = \frac{174815250030533}{310875000000000000} > 0.$$

All vector images are evaluated on the support enlarged by two sites at each end, so no outgoing coordinate is omitted. The full vectors and image coordinates are placed in the supplement because they are direct integer reconstructions of the six rows above.

5.3 Uniform control of the large gaps

One fixed vector covers the infinite tail. On the consecutive interval $[-2, 11]$, take

$$v_* = (4, 0, 7, 8, 8, 9, 1, 7, 3, 6, 1, 4, 1, 2), \qquad \|v_*\|^2 = 391.$$

Only coefficients $\tau_i^{(g)}$ with $-4 \le i \le 11$ enter the full image on $[-4, 13]$. The right interface endpoint can therefore affect the calculation only when $g = 9$ or $g = 10$; every $g \ge 11$ has the same local coefficient word. Direct use of the four-term formula for A_g gives

$$\|A_9 v_*\|^2 = 3102, \qquad \|A_{10} v_*\|^2 = 3094, \qquad \|A_g v_*\|^2 = 3094 \quad (g \ge 11).$$

Consequently every $g \ge 9$ satisfies

$$E(g) \ge \frac{3094}{391} = \frac{182}{23}.$$

The remaining comparison is again exact:

$$\frac{182}{23} - T = \frac{10563229091557}{2875000000000000} > 0.$$

This proves the required strict separation uniformly on the entire large-gap tail. The three displayed squared norms, together with finite propagation, are a proof for all $g \ge 9$, not a sample of those gaps.

5.4 Reference and single-gap hierarchy

Theorem 5.1 (Complete abnormal single-gap hierarchy)**.** *For both τ-lifts and both interface orientations,*

$$E(4) = \eta < c_6 = E(6),$$

and, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

Thus G6 uniquely minimizes the squared spectral edge among abnormal positive single-gap interfaces.

Proof. The reference and G6 equalities come from Proposition 2.3 and equation (6); the strict inequality $\eta < c_6$ follows from the exact isolating intervals already displayed. Subsection 5.2 treats every positive gap below nine except 4 and 6, and Subsection 5.3 treats every gap at least nine. These sets exhaust the remaining positive integers. Lift negation preserves the squared spectrum, and reflection transfers the result to the opposite orientation. $\square$

The theorem compares single abnormal positive gaps only. It makes no claim about arbitrary multi-gap configurations, interacting interfaces, or general finite cores. The constant $1/250$ is a convenient certified uniform separation; no optimality is claimed for the size of that margin.

6 Phase Slips on Finite Rings

6.1 Legal residue constructions

We now close one, two, or three G6 interfaces on finite rings. Exponent notation 4^a denotes a consecutive reference gaps.

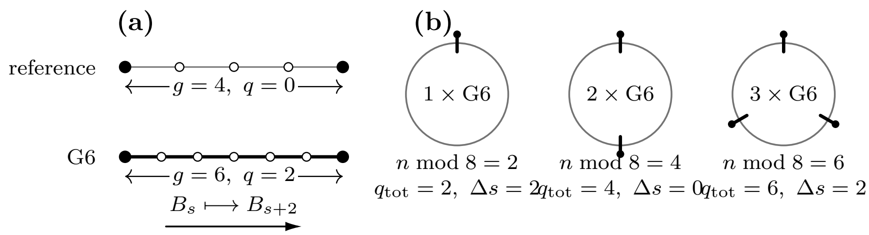

Figure 2: Reference gaps and charge-compatible phase slips. A G6 replaces $g = 4, q = 0$ by $g = 6, q = 2$; one, two, and three copies provide the mod-eight total charges for residues $2, 4, 6$, while their mod-four sector transport is checked independently.

For $n = 8k + 2$, use

$$\mathcal{G}_2(k) = [6, 4^{2k-1}].$$

Its gap sum is $6 + 4(2k - 1) = 8k + 2$, it has $2k$ positive Q-sites, and its single interface has return distance $D_2(n) = n$.

For $n = 8k + 4$, use

$$\mathcal{G}_4(k) = [6, 4^{k-1}, 6, 4^{k-1}].$$

Its sum is $12 + 8(k - 1) = 8k + 4$, it again has $2k$ positive sites, and the two interface cores are separated by $D_4(n) = n/2$ in both directions.

For $n = 8k + 6$, put

$$a = \left\lfloor \frac{2k-3}{3} \right\rfloor, \qquad b = \left\lfloor \frac{2k-2}{3} \right\rfloor, \qquad c = \left\lfloor \frac{2k-1}{3} \right\rfloor$$

and use

$$\mathcal{G}_6(k) = [6, 4^a, 6, 4^b, 6, 4^c].$$

Writing $2k - 3 = 3u + v$, $v \in \{0, 1, 2\}$, verifies $a + b + c = 2k - 3$. Hence the gap sum is

$$18 + 4(a + b + c) = 8k + 6,$$

there are $2k$ positive sites, and the minimum core separation is

$$D_6(n) = 6 + 4 \left\lfloor \frac{2k-3}{3} \right\rfloor.$$

These formulas are used only when their exponents are nonnegative.

In every construction the number of positive Q-sites and n are even. Thus

$$\prod_i Q_i = (-1)^{n-2k} = 1,$$

so both cyclic τ-lifts exist. The mod-eight charge check is separate: each G6 contributes $q = 2$, and the total charges $2, 4, 6$ agree with the three order residues. The translation-sector check uses Proposition 3.2: the total shifts are $2, 0, 2$ modulo four, equal respectively to n modulo four. After these two independent closure checks, either Hamilton holonomy $\alpha = \pm 1$ may be chosen.

Figure 2 separates the mod-eight charge bookkeeping from the mod-four sector transport in these three families.

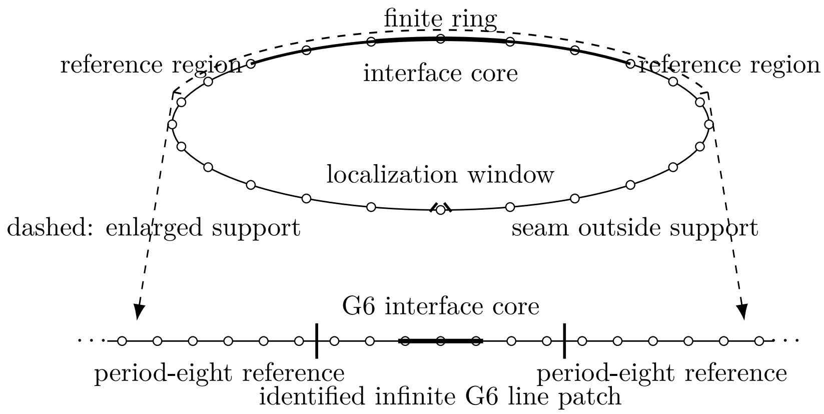

Figure 3: Local-to-global identification. A finite-ring cutoff and its range-four enlargement see either reference coefficients or one interface core; after unwrapping and switching, the latter is exactly a finite section of the bilateral G6 model. The holonomy seam is placed outside the interface support.

6.2 Identification of local patches

Consider any of the preceding rings with $r \in \{1, 2, 3\}$ marked G6 cores, and let D be their minimum cyclic site separation; for $r = 1$, set $D = n$. Here D is measured in ring sites, not period-eight cells. Let a real cutoff be supported in a cyclic interval of radius R. Since $H = A^2$ has range four, its enlarged support has radius $R + 4$.

Proposition 6.1 (Finite-ring patch identification)**.** *Assume*

$$2(R + 4) < D.$$

Every enlarged localization patch is, after cyclic translation, optional reflection, and diagonal switching, a finite section of either the period-eight reference operator or the canonical bilateral G6 operator. This covers both τ-lifts, both interface orientations, both Hamilton holonomies, patches crossing the displayed seam, and cyclic wraparound.

Proof. The separation inequality ensures that an enlarged support meets at most one non-four gap. If it meets none, its positive Q-sites form one congruence class modulo four, so translation identifies the coefficient word with a reference sector B_s. If it meets one, translate the left endpoint of the gap to zero. The visible gaps are then four on both sides of one gap six, hence the word is the forward G6 model; reversing the cyclic coordinate gives the other orientation.

Each enlarged support is a proper arc. On such an arc, recursive diagonal switching makes every step-one edge positive. Two lifts of the same local Q-word differ by a common sign, removed after squaring by equation (2). The Hamilton holonomy can be placed on any chosen step-one seam. Move it outside every interface support; if a bulk patch crosses the initially displayed seam, the same proper-arc switching moves that sign outside the local compression. Unwrapping a proper arc before translation handles cyclic wraparound. These operations prove equality of all coefficients needed by the range-four quadratic form, not merely similarity of a central subword. □

The unwrapping and seam placement in the proof are illustrated in Figure 3.

6.3 Discrete IMS localization

Let H be self-adjoint and let χ_j be real diagonal matrices with $\sum_j \chi_j^2 = I$. Entrywise,

$$[\chi, [\chi, H]]_{ab} = (\chi(a) - \chi(b))^2 H_{ab}.$$

For each pair a, b, the coefficient of H_{ab} in the following expression is one. Thus the exact discrete IMS identity is

$$H = \sum_j \chi_j H \chi_j + \frac{1}{2} \sum_j [\chi_j, [\chi_j, H]]. \tag{9}$$

For $R \geq 4$, define on the cycle

$$f_R(k) = \max\left(0, 1 - \frac{\mathrm{dist}(k, 0)}{R}\right), \qquad C_R = \sum_k f_R(k)^2 = \frac{2R^2 + 1}{3R},$$

and use every cyclic translate

$$\chi_j(a) = \frac{f_R(a - j)}{\sqrt{C_R}}.$$

If $n > 2R + 4$, the supports do not self-collide in a range-four calculation, and $\sum_j \chi_j(a)^2 = 1$. For cyclic distance $d \leq 4$, direct summation gives

$$S_d(R) := \sum_j (\chi_j(a) - \chi_j(b))^2 = \frac{3(2d^2 R - d(d^2 - 1))}{R(2R^2 + 1)}.$$

From equation (3), the absolute off-diagonal weights of H at distances $1, 2, 3, 4$ are bounded by $2, 1, 2, 1$. The following Schur row bound is a uniform operator-norm estimate for the IMS remainder:

$$2S_1 + S_2 + 2S_3 + S_4 = \frac{240R - 342}{R(2R^2 + 1)} \leq \frac{120}{R^2}. \tag{10}$$

Proposition 6.2 (Separated-interface cap)**.** *If $2(R + 4) < D$, then every legal ring above satisfies*

$$\rho(A)^2 \leq c_6 + \frac{240R - 342}{R(2R^2 + 1)} \leq c_6 + \frac{120}{R^2}.$$

Proof. By Proposition 6.1, every localized vector is unitarily identified with a reference or G6 line vector. Their squared spectral edges are $\eta < c_6$ and c_6, respectively, so

$$\langle \chi_j x, H \chi_j x \rangle \leq c_6 \|\chi_j x\|^2.$$

Sum over j, use $\sum_j \chi_j^2 = I$, and apply equations (9)–(10). Taking the supremum over unit vectors controls the full finite-ring spectral top. ☐

6.4 Residue upper bounds and the analytic tail

For $s \in \{2, 4, 6\}$, the corresponding family has a fixed number of interfaces while $D_s(8k + s) \to \infty$. Choose

$$R = \left\lfloor \frac{D_s(8k + s) - 9}{2} \right\rfloor.$$

Then $2(R + 4) < D_s(8k + s)$, and $R \to \infty$. Proposition 6.2 gives the one-sided conclusion

$$\limsup_{k \to \infty} m_{8k+s}^2 \leq c_6, \qquad s \in \{2, 4, 6\}. \tag{11}$$

This proves neither a lower bound nor a limit or common liminf.

We next obtain an explicit analytic tail. The comparison threshold obeys

$$\theta_n > 8 - \frac{200}{n^2},$$

by $\cos x > 1 - x^2/2$ and $\pi^2 < 10$. Let

$$\mathcal{E}(R) = \frac{240R - 342}{R(2R^2 + 1)}.$$

At the first order in each residue class at or above 240, exact arithmetic gives:

n	certified squared cap	$8 - 200/n^2$
240	1561/200	2303/288
242	257368059342729114019/32519875000000000000	117078/14641
244	14532080076773342617/1829625000000000000	59511/7442
246	2591140328128938813/324125000000000000	120982/15129

In every row the middle fraction is strictly smaller than the last by integer cross-multiplication. The first row uses the period-eight competitor; the other rows are the upper endpoint for c_6 plus $\mathcal{E}(R)$ for the displayed residue construction.

For $R \geq 4$,

$$\mathcal{E}(R) - \mathcal{E}(R+1) = \frac{6(160R^3 - 102R^2 - 262R - 171)}{R(R+1)(2R^2+1)(2R^2+4R+3)} > 0.$$

The numerator equals 7389 at $R = 4$, and its derivative is positive thereafter. Thus the strict decrease is analytic, not numerical. For $s \in \{2, 4, 6\}$, D_s and the chosen R do not decrease, so the spectral cap does not increase, whereas $8 - 200/n^2$ strictly increases. For residue zero, the period-eight cap 1561/200 is constant and the same threshold lower bound increases. The four endpoint checks therefore propagate to every later order. We have proved

$$m_n < \rho_-(n) \qquad \text{for every even } n \geq 240.$$

This phase-slip argument explains why failure is eventually uninterrupted. It does not locate the sharp beginning at 48; that conclusion also uses the exact finite bridge in Section 7. Neither the exact-$2r$ count nor the rank-two multiplicity is used in this analytic tail.

6.5 Structural refinement for separated interfaces

Theorem 6.3 (Separated-interface cluster)**.** *Let a legal finite ring contain exactly $r \in \{1, 2, 3\}$ separated G6 interfaces, with either τ-lift, either orientations, and either Hamilton holonomy. If the minimum cyclic core separation satisfies $D \geq 1040$, then*

$$\operatorname{rank} \mathbf{1}_{[c_6 - 1/400,\, c_6 + 1/400]}(H) = 2r.$$

Eigenvalues are counted with multiplicity; no individual simplicity is asserted.

Each isolated G6 has a rank-two squared eigenspace, one mode from each simple unsquared partner. Truncating both modes at each interface gives $2r$ quasimodes. Exact tail and Gram bounds keep them independent and give the lower spectral count. On their orthogonal complement, patch identification, the single-interface complement bound, and discrete IMS give the matching upper count. The full Gram, complement, and effective-operator estimates belong to the separate supplement. Theorem 6.3 is a structural refinement of the separated-interface picture. It is not used in Proposition 6.2, the analytic $n \geq 240$ tail, or the exact argument placing the onset of continuous failure at 48.

7 Finite Completion of the Classification

7.1 Exact-computation protocol

The finite part of the proof follows one contract throughout:

$$\begin{aligned}\text{mathematical reduction} &\longrightarrow \text{finite exact object}\\ &\longrightarrow \text{independent machine verification} \longrightarrow \text{mathematical consequence.}\end{aligned}$$

These are logical stages, not numerical approximations.

For a signed adjacency matrix A,

$$\rho(A)^2 = \lambda_{\max}(A^2) = \max_{v \neq 0} \frac{\|Av\|^2}{\|v\|^2}.$$

Thus an exact integer vector with quotient at least θ_n proves the required lower bound for that class. In the other direction, if positive integers p, q satisfy

$$pI - qA^2 \succ 0,$$

then $\rho(A)^2 < p/q$. A strict exact comparison $p/q < \theta_n$ therefore certifies a counterexample. Positive definiteness is established by fraction-free minors or rational LDL^{T} pivots, and algebraic thresholds are selected by exact root isolation. Floating-point calculations may suggest a vector or an interval, but they make no accepting decision.

The main text proves that the finite domains below are exhaustive and explains why their exact terminal decisions imply the spectral statements. Appendix B supplies the full mathematical finite-object definitions and completeness proofs; removable reconstruction records belong to the supplement.

7.2 Orders 8 through 32

Proposition 7.1. *For every even $8 \leq n \leq 30$,*

$$\rho(A_\sigma)^2 \geq \theta_n \qquad \textit{for every signing } \sigma.$$

Consequently $m_n = \rho_-(n)$ at these orders.

Proof. Hamilton gauge reduces every signing to a finite switching quotient. For $8 \leq n \leq 20$, every switching class is represented directly. At $n = 22$, the reduced (Q, α) quotient is exhausted, with the two τ_0-lifts covered by their identical squared spectra. For $n = 24, 26, 28, 30$, fixed-weight cyclic words followed by dihedral canonicalization represent the same full quotient without duplication. Lift negation preserves the squared spectrum, and both holonomies remain in the domain.

Every nonthreshold class has a primitive integer vector whose exact Rayleigh quotient is at least θ_n. Each remaining threshold class is closed by exact characteristic-polynomial equality at the largest squared root. An independent reconstruction verifies the represented quotient and repeats every terminal spectral decision using integer and algebraic arithmetic. Hence the universal lower bound holds. Proposition 2.2 supplies the reverse inequality $m_n \leq \rho_-(n)$, proving equality. $\square$

The next order fails. Take $\alpha = +1$, all displayed distance-one signs positive, and the period-eight triangle-flux word

$$\tau = (+, +, -, +, -, -, +, -)^4.$$

Let A_{32} be the resulting exact signed matrix. Bareiss minors and an independent rational LDL^{T} decomposition give

$$1561I_{32} - 200A_{32}^2 \succ 0.$$

Therefore

$$\rho(A_{32})^2 < \frac{1561}{200} < \frac{11896117236720419}{1523321182060814} < \theta_{32},$$

where the middle fraction is an exact lower isolating endpoint for θ_{32}. Thus $m_{32} < \rho_-(32)$. Together with Proposition 7.1, this proves that 32 is the first failing order.

7.3 Orders 34 through 46

For $n \in \{34, 36, 38, 42, 44, 46\}$, a local finite-state reduction proves the universal lower bound. We give its completeness argument because the computation is meaningful only after this reduction.

Fix a support $S = \{0, \dots, L-1\}$. A length-$(L+1)$ local Q-window W determines the exact integer map

$$C_W : \mathbb{R}^L \longrightarrow \mathbb{R}^{L+4}, \qquad M_W = C_W^{\mathsf{T}} C_W = P_S A^2 P_S.$$

For a strict rational upper isolating endpoint $b_n > \theta_n$, an integral vector satisfying

$$v^{\mathsf{T}} M_W v > b_n\, v^{\mathsf{T}} v$$

excludes every global signing containing W. Exact local Rayleigh certificates close the excluded side. Independently, fraction-free Sylvester tests decide every window:

$$a_n I - M_W \succ 0 \Longrightarrow \lambda_{\max}(M_W) < a_n < \theta_n,$$

$$b_n I - M_W \not\succ 0 \Longrightarrow \lambda_{\max}(M_W) \geq b_n > \theta_n.$$

The windows satisfying the first condition form the surviving set, and no local window remains undecided.

Build the overlap graph whose edges are the surviving length-$(L+1)$ words and whose vertices are their length-L prefixes and suffixes. If the appended bit records $Q_i = +1$, lift each vertex to (s, ε), $\varepsilon \in \mathbb{Z}/2\mathbb{Z}$, and update

$$(s, \varepsilon) \longmapsto (s', \varepsilon + b_L).$$

A length-n closed walk gives a cyclic word because consecutive edges overlap, the base-state closure enforces the wraparound windows, and parity closure says $\prod_i Q_i = 1$. Conversely, reading any liftable cyclic Q-word with no excluded window produces such a closed walk: all windows are surviving edges, cyclicity closes the base state, and liftability closes parity. Hence the remaining global candidates are exactly the parity-even closed walks, not a sampled subset.

Rotation and reflection preserve the squared spectrum, so dihedral reduction removes duplicates without losing a class. Both holonomies are then checked for every canonical Q-word. The exact terminal counts are

n	34	36	38	42	44	46
canonical Q-classes	1	1	3	7	10	10
terminal (Q, α)-records	2	2	6	14	20	20

and total

$$2(1 + 1 + 3 + 7 + 10 + 10) = 64.$$

Every terminal is closed by either exact threshold-factor equality or a strict integral Rayleigh inequality. Independent reconstruction confirms all 64 decisions, and no terminal remains unresolved. It follows that every signing satisfies

$$\rho(A_\sigma)^2 \geq \theta_n \qquad (n = 34, 36, 38, 42, 44, 46).$$

Candidate attainment then proves

$$m_n = \rho_-(n) \qquad (n = 34, 36, 38, 42, 44, 46).$$

Order 40. This order is logically separate from the six lower-bound decisions. Use

$$Q = 1000100010001000100010001000100010001000, \qquad \alpha = -1,$$

where 1 encodes $Q_i = +1$ and 0 encodes $Q_i = -1$. With $\tau_0 = 1$, reconstruct A_{40} in Hamilton gauge. Exact rational elimination gives forty positive pivots in

$$15541 I_{40} - 2000 A_{40}^2,$$

so

$$\rho(A_{40})^2 < \frac{15541}{2000} < \frac{63}{8} = 8 - \frac{200}{40^2} < \theta_{40}.$$

The first strict difference is 209/2000, and the last inequality is the elementary cosine bound used in Section 6.4. Thus $m_{40} < \rho_-(40)$.

7.4 The exact bridge from 48 to 238

The even interval $48 \leq n < 240$ contains exactly

$$\frac{238 - 48}{2} + 1 = 96$$

orders. At each order use the deterministic signing family

n mod 8	finite signing
0	period-eight repetition
2	one G6 phase slip
4	two G6 phase slips
6	three G6 phase slips

with holonomy -1 for $n \equiv 0 \pmod 4$ and $+1$ for $n \equiv 2 \pmod 4$. These are the same explicit residue constructions as in Section 6; no existence choice is hidden.

For every one of the 96 matrices, an exact rational number t_n satisfies

$$t_n I - A_n^2 \succ 0, \qquad t_n < 8 - \frac{200}{n^2}.$$

The finite object is the complete order-indexed list of the explicit signing and its rational positive-definiteness certificate. An independent verification rebuilds each full signed matrix and repeats exact sparse rational elimination in a different ordering. Positive definiteness and the cosine bound imply

$$\rho(A_n)^2 < t_n < 8 - \frac{200}{n^2} < \theta_n.$$

Hence every even order from 48 through 238 fails. This exact finite verification, joined to the analytic $n \geq 240$ tail, moves the beginning of continuous failure from an eventual statement to the sharp order 48.

7.5 Exhaustive synthesis

Orders	Result	Proof mechanism
even 8–30	equality	exact exhaustion plus candidate attainment
32	failure	exact positive-definiteness witness
$34, 36, 38, 42, 44, 46$	equality	finite-state closure plus attainment
40	failure	exact rational LDL^{T} witness
even 48–238	failure	96 exact rational certificates
even $n \geq 240$	failure	phase slips, patch identification, and IMS

Proof of Theorem 1.1. The even integers $n \geq 8$ are the disjoint union

$$\{8, 10, \ldots, 30\}, \quad \{32\}, \quad \{34, 36, 38, 42, 44, 46\}, \quad \{40\},$$
$$\{48, 50, \ldots, 238\}, \quad \{240, 242, 244, \ldots\}.$$

Proposition 7.1 and the finite-state closure give a universal lower bound on the first and third sets. Candidate attainment gives the reverse inequality there, hence equality. The exact witnesses at 32 and 40, the 96-order finite bridge, and the analytic tail give strict counterexamples on the other four sets. Therefore

$$m_n < \rho_-(n) \quad \iff \quad n = 32, \quad n = 40, \quad \text{or} \quad n \geq 48$$

for even $n \geq 8$, and equality holds exactly on the validity set stated in Theorem 1.1. □

The finite classification determines the irregular small-order pattern. The phase-slip argument explains eventual uninterrupted failure. The sharp continuous onset 48 is a consequence of the exact finite bridge together with the analytic tail; neither component alone supplies that conclusion.

8 Concluding Remarks

We have determined exactly when the distinguished twisted signing minimizes the adjacency spectral radius of the signed square of an even cycle. The answer is not monotone in the order: equality holds at every even order $n \geq 8$ except $n = 32$, $n = 40$, and all even $n \geq 48$. The proof also identifies a structural mechanism behind the eventual failure. A period-eight reference phase admits an elementary six-gap interface whose squared spectral edge is c_6, and separated copies of this interface can be placed on large finite rings. This mechanism proves the analytic tail, while exact finite arguments determine the exceptional orders and the onset at 48.

The result leaves three natural questions. First, along each nonzero even residue class modulo 8, does the sequence of minimum squared spectral radii converge to c_6? The present argument proves only the corresponding upper limit bound, and supplies no matching lower bound.

Second, is the six-gap interface optimal in a broader class of sector-changing finite cores? Our comparison theorem is restricted to single-gap interfaces, so it does not exclude a genuinely multi-gap core with a smaller spectral edge.

Third, do spectral-radius minimizers at sufficiently large orders have an asymptotic description in terms of well-separated elementary phase slips? The constructions used here establish effective competitors, but they do not classify the minimizing switching classes or force such a decomposition.

Data and Code Availability

Repository, source-checkpoint, and archive details are omitted from this anonymous review copy. The identified submission will include the complete development-repository and reproducibility statement.

A Exact Spectral Certification of the Reference Phase and G6

This appendix supplies the exact algebra behind Proposition 2.3, Lemma 4.2, and Proposition 4.3. The order of the argument is important. We first construct a genuine bilateral eigenfunction by an unsquared stable–unstable matching condition. Only after physical existence has been proved do we use elimination to identify its squared energy. The same unsquared condition is then used to reject every additional algebraic candidate. Thus no zero of a resultant is treated as an eigenvalue merely because it survives elimination.

Throughout, all interval endpoints are rational. An assertion that a polynomial or a radical expression has a fixed sign on an interval means that repeated squaring, with the sign of each unsquared side retained, gives the displayed rational enclosure. Sturm counts use the ordinary signed remainder sequence over $\mathbb{Q}$.

A.1 The reference Floquet calculation

The reference lift is

$$\tau_{\mathrm{ref}} = (1, 1, -1, 1, -1, -1, 1, -1).$$

For $i = 8m + r$, put $u_i = z^m v_r$. In the ordered basis $(v_0, \dots, v_7)$, the unsquared Bloch fiber is

$$A_{\mathrm{ref}}(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & z^{-1} & z^{-1} \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} \\ 1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & -1 \\ z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & -z & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}.$$

Reversing every undirected transition gives $A_{\mathrm{ref}}(z)^{\mathsf{T}} = A_{\mathrm{ref}}(z^{-1})$, so the fiber is Hermitian for $|z| = 1$. Fraction-free expansion gives

$$\begin{aligned} \det(xI - A_{\mathrm{ref}}(z)) = {} & x^8 - 16x^6 + (80 - 2(z + z^{-1}))x^4 \\ & + (-128 + 16(z + z^{-1}))x^2 + (z^2 + z^{-2}) - 13(z + z^{-1}) + 40. \end{aligned} \tag{12}$$

With $y = x^2$ and $c = z + z^{-1}$, this becomes

$$P(y, c) = y^4 - 16y^3 + (80 - 2c)y^2 + (-128 + 16c)y + c^2 - 13c + 38. \tag{13}$$

On the unit circle, $c \in [-2, 2]$. At $c = 2$, translation by four in the spectral variable gives

$$P(X + 4, 2) = X^4 - 20X^2 + 80.$$

Consequently the upper squared edge at $z = 1$ is

$$\eta = 4 + s, \qquad s = \sqrt{10 + 2\sqrt{5}}.$$

To prove that no other fiber reaches it, put $u = y - \eta$ and $t = 2 - c$. Direct substitution in (13) yields

$$\begin{aligned} P(\eta + u, 2 - t) = {} & u^4 + 4su^3 + 2u^2 t + (40 + 12\sqrt{5})u^2 + 4sut \\ & + 8\sqrt{5}\, su + t^2 + (4\sqrt{5} - 3)t. \end{aligned} \tag{14}$$

Every coefficient is positive. For $u, t \geq 0$, the right-hand side vanishes only at $(u, t) = (0, 0)$. Hence $y \leq \eta$ on every unit-circle fiber, and equality forces $c = 2$, hence $z = 1$. This proves both the value and the uniqueness assertion in Proposition 2.3. It also gives the useful strict bound $\eta < 391/50 < 8$.

The same algebra appears in transfer form. For a general lift τ, write

$$X_i = (u_{i+1}, u_i, u_{i-1}, u_{i-2})^{\mathsf{T}}.$$

The equation $A_\tau u = \lambda u$ is equivalent to

$$X_{i+1} = T_i(\lambda)X_i, \qquad T_i(\lambda) = \begin{pmatrix} -\tau_i & \tau_i\lambda & -\tau_i & -\tau_i\tau_{i-2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{15}$$

Each T_i is invertible. For one reference cell set $M_8(\lambda) = T_7(\lambda) \cdots T_0(\lambda)$. Exact multiplication gives, with $y = \lambda^2$,

$$\det(zI - M_8(\lambda)) = z^4 + a(y)z^3 + b(y)z^2 + a(y)z + 1, \tag{16}$$

where

$$a(y) = -2y^2 + 16y - 13, \qquad b(y) = y^4 - 16y^3 + 80y^2 - 128y + 40. \tag{17}$$

After division by z^2, the substitution $w = z + z^{-1}$ reduces the quartic to

$$f(w; y) = w^2 + a(y)w + b(y) - 2 = 0. \tag{18}$$

Its discriminant is $-12y^2 + 96y + 17$, and the reciprocal multiplier pairs are obtained from

$$w_\mp(y) = \frac{2y^2 - 16y + 13 \mp \sqrt{-12y^2 + 96y + 17}}{2}, \qquad z_{s,u}(w) = \frac{w \mp \sqrt{w^2 - 4}}{2}. \tag{19}$$

Equation (14) excludes a unit-circle multiplier for every $y > \eta$. Reciprocity and continuity then give two multipliers in $|z| < 1$ and two in $|z| > 1$, counted algebraically. The complete multiplier discriminant factors as

$$\begin{aligned} &(y^2 - 12y + 34)(y^2 - 4y + 2)(12y^2 - 96y - 17)^2 \\ &\qquad \cdot (y^4 - 16y^3 + 76y^2 - 96y + 16). \end{aligned} \tag{20}$$

Above the upper endpoint of J_6, defined below, the only collision is

$$y_* = 4 + \frac{\sqrt{627}}{6}, \qquad w = \frac{95}{12} > 2. \tag{21}$$

Indeed, the positive roots of the first two quadratic factors in (20) satisfy

$$6 + \sqrt{2} < \frac{15}{2} < a_6, \qquad 2 + \sqrt{2} < \frac{7}{2} < a_6,$$

and every root of the final quartic is at most $\eta < a_6$ by the reference-edge calculation. The factor $12y^2 - 96y - 17$ has one negative root and the single positive root y_*. This proves directly, without a floating root comparison, that y_* is the unique collision on $[a_6, 16]$. It is therefore a collision inside the hyperbolic region, not a loss of the stable–unstable splitting.

A.2 Exact transfer through the G6 core

For the forward interface, take

$$Q_i = 1 \iff \big(i \le 0,\ i \equiv 0 \pmod 4\big) \text{ or } \big(i \ge 6,\ i \equiv 6 \pmod 4\big),$$

fix $\tau_0 = 1$, and impose $\tau_{i+1} = Q_i \tau_i$. The left and right cuts are -8 and 14. Thus

$$D_6(\lambda) = T_{13}(\lambda) \cdots T_{-8}(\lambda), \qquad \det D_6(\lambda) = 1. \tag{22}$$

Equations (15) and (22), together with the displayed definition of Q, are an exact finite specification of every entry of the defect transfer; each entry is an integer polynomial in λ of degree at most eleven. The two bulk monodromies at these cuts are

$$M_L = T_{-9} \cdots T_{-16}, \qquad M_R = T_{21} \cdots T_{14}.$$

They are transfer representatives of the translated reference phases B_0 and B_2.

For $y > \eta$ and $\lambda = \sqrt{y} > 0$, let $U_L(\lambda)$ be the two-dimensional space of left-cut states decaying under backward iteration, and let $S_R(\lambda)$ be the two-dimensional space of right-cut states decaying under forward iteration. Since the recurrence is invertible, a nonzero bilateral ℓ^2 solution exists if and only if

$$D_6(\lambda) U_L(\lambda) \cap S_R(\lambda) \neq \{0\}. \tag{23}$$

If u_1, u_2 and s_1, s_2 are bases of these planes, define

$$\mathcal{E}_6(y) = \det[D_6(\sqrt{y})u_1, D_6(\sqrt{y})u_2, s_1, s_2]. \tag{24}$$

A change of basis multiplies $\mathcal{E}_6$ by a nowhere-zero factor. Thus its zero set is precisely the coordinate-free condition (23). In particular, changing cuts by full periods or changing Grassmann charts cannot create or remove a physical zero.

For the signed interval calculations below we fix the normalization completely. Order the two stable multipliers by

$$0 < z_1 < z_2 < 1$$

and the unstable multipliers as z_1^{-1}, z_2^{-1}. For a row set $a < b < c$, define the cofactor vector by

$$\big(v_{abc}(z)\big)_j = (-1)^j \det\big((M - zI)_{\{a,b,c\},\,\{0,1,2,3\}\setminus\{j\}}\big).$$

Use rows 012, divide each two-vector wedge by the ordered Vandermonde $z_1 - z_2$, and order the four columns in (24) as

$$\big[D_6 u(z_1^{-1}), D_6 u(z_2^{-1}), s(z_1), s(z_2)\big].$$

This convention fixes a continuous orientation on J_6. The endpoint and derivative signs below refer to this normalized determinant, rather than to an arbitrary basis representative of the same zero set.

The essential-spectrum assertion used here can also be seen directly. Cutting $H_6 = A_6^2$ on both sides of its finite core changes it by finite rank into two periodic half-line compressions and a finite block. A cutoff Bloch wave proves that the spectrum of a whole-line periodic operator lies in the essential spectrum of its half-line compression. Conversely, if $R = (H_{\text{per}} - x)^{-1}$, then the compressed operator PRP is a two-sided inverse modulo finite-rank cross-boundary terms. Hence the half-line compression is Fredholm whenever x is outside the whole-line spectrum. It follows that

$$\sigma_{\text{ess}}(H_6) = \sigma(H_{\text{ref}}), \qquad \sup \sigma_{\text{ess}}(H_6) = \eta.$$

For $y > \eta$, the hyperbolic splitting above shows that every matched state decays exponentially. This proves that (23) is both necessary and sufficient for a discrete physical level.

A.3 Physical realization and algebraic identification

Put

$$a_6 = \frac{7905369311620327}{10^{15}}, \qquad b_6 = \frac{7905369311620328}{10^{15}} = \frac{988171163952541}{125000000000000}, \qquad J_6 = [a_6, b_6].$$

On J_6, the four multipliers are distinct and positive, with two on each side of the unit circle. Cofactor eigenvectors formed from rows $0, 1, 2$ of $M - zI$ have a nonzero component throughout this interval, so they define a valid normalization of (24). Exact outward evaluation gives

$$\begin{aligned}
-\frac{1137363381850112739610 51}{10^{30}} &\le \mathcal{E}_6(a_6), \\
\mathcal{E}_6(a_6) &\le -\frac{227472676370022547922 1}{2 \cdot 10^{28}} < 0, \qquad (25) \\
0 < \frac{469153221168381793862160 7}{5 \cdot 10^{29}} &\le \mathcal{E}_6(b_6), \\
\mathcal{E}_6(b_6) &\le \frac{187661288467352717544864 3}{2 \cdot 10^{29}}, \qquad (26) \\
0 < \frac{4748398569130503440680129312442433244027}{5 \cdot 10^{29}} &\le \mathcal{E}_6'(y), \\
\mathcal{E}_6'(y) &\le \frac{1187100548105535925201118811466127038887}{12500000000000000000000000000000} \quad (y \in J_6). \qquad (27)
\end{aligned}$$

The derivative is with respect to y. Equations (25)–(27) prove, by the intermediate-value theorem and strict monotonicity, that there is exactly one positive-branch physical zero in J_6. Since $y > 0$, it is also a simple zero as a function of λ. If the matching intersection had dimension at least two, the matching matrix in (24) would have rank at most two; its adjugate and hence the first derivative of its determinant would vanish. Therefore the positive eigenspace is one-dimensional.

It remains to identify the squared energy. Let z_1, z_2 be the stable multipliers and put

$$S = z_1 + z_2, \qquad P = z_1 z_2.$$

The reciprocal quartic (16) gives

$$S(P+1) + a(y)P = 0, \qquad P^2 + S^2 + 1 - b(y)P = 0. \tag{28}$$

After dividing the stable wedge by the nonzero Vandermonde $z_1 - z_2$, every Plücker coordinate is symmetric in z_1, z_2 and hence polynomial in S, P. Substituting the first relation in the second gives the product equation

$$R(P, y) := (P^2 + 1 - b(y)P)(P+1)^2 + a(y)^2 P^2 = 0. \tag{29}$$

After removal of the proved nonzero chart denominators, the genuine unsquared matching numerator has the form

$$E_0(y, P) + \lambda E_1(y, P).$$

This is also a finite exact specification of E_0, E_1: form the matrices T_i in (15), multiply them in the order (22), form the signed cofactor vectors just defined, divide the wedges by their ordered Vandermonde factors, and reduce the resulting symmetric polynomials by (28). No fitted coefficient or numerical interpolation enters this construction. Thus every physical zero annihilates

$$G(P, y) = E_0(y, P)^2 - y E_1(y, P)^2. \tag{30}$$

The resultant of R and G with respect to P has degree 108 in y. Up to a nonzero rational constant, its exact factorization is

$$r_2(y)^{16} r_{4,1}(y)^2 r_{4,2}(y)^2 p_6(y)^2 q_6(y)^4, \tag{31}$$

where

$$\begin{aligned}
r_2(y) &= 2y^2 - 16y + 13, \\
r_{4,1}(y) &= y^4 - 16y^3 + 68y^2 - 32y + 52, \\
r_{4,2}(y) &= 9y^4 - 144y^3 + 708y^2 - 1056y + 404, \\
p_6(y) &= 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\
&\quad - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\
&\quad - 19968y + 256, \\
q_6(y) &= 16y^{10} - 484y^9 + 5796y^8 - 34617y^7 + 105000y^6 \\
&\quad - 135904y^5 + 7632y^4 + 60620y^3 + 57232y^2 - 52880y + 64.
\end{aligned}$$

Let $V_f(x)$ denote the number of sign changes in the Sturm sequence of f at x. The exact endpoint variations are

factor	$V_f(a_6)$	$V_f(b_6)$	$V_f(16)$
r_2	0	0	0
$r_{4,1}$	2	2	2
$r_{4,2}$	0	0	0
p_6	3	2	1
q_6	2	2	1

In particular, p_6 has exactly one root in J_6, while every other factor in (31) has none there. Since a physical zero was established before the elimination, its square is the unique root $c_6 \in J_6$ of p_6. This proves the algebraic identification in Lemma 4.2 without reversing the logical implication from a resultant candidate to a physical state.

A.4 A complete Grassmann atlas above c_6

We now cover

$$I = [b_6, 16].$$

The stable plane can be continued through the collision (21) without choosing the two multipliers separately. Indeed, if $t = P + P^{-1}$, then

$$(t+2)(t - b(y)) + a(y)^2 = 0. \tag{32}$$

The larger real solution $t = t_+(y) > 2$ selects the physical stable branch by

$$P = \frac{2}{t_+(y) + \sqrt{t_+(y)^2 - 4}}, \qquad S = -\frac{a(y)P}{P+1}. \tag{33}$$

The unstable plane uses the reciprocal multipliers.

For a cofactor row set abc, let $v_{abc}(z)$ be the signed vector of the 3×3 minors of the rows a, b, c of $M - zI$. Then

$$\Omega_{abc}(z_1, z_2) = \frac{v_{abc}(z_1) \wedge v_{abc}(z_2)}{z_1 - z_2}$$

is symmetric and extends at $z_1 = z_2 = z$ to $v_{abc}(z) \wedge \partial_z v_{abc}(z)$. This is the confluent wedge used at y_*. A nonzero Plücker coordinate of Ω_{abc} gives an affine chart for the physical plane.

The following three closed charts cover I:

interval	cofactor rows	right stable minor	left unstable minor
$[b_6, 31663/4000]$	013	p_{23}	p_{01}
$[79157/10000, 79159/10000]$	012	p_{01}	p_{02}
$[158317/20000, 16]$	013	p_{23}	p_{01}

For rows 013, elimination of either selected section against (29) shows that its only possible zero in I comes from

$$h(y) = 3y^2 - 24y + 2.$$

Its Sturm variations at $7915780041490243/10^{15}$ and $7915780041490244/10^{15}$ are respectively one and zero. Hence its unique root lies strictly inside the middle chart and outside both outer charts. On the middle interval, the two selected sections satisfy

$$0 < \frac{372155273746344547888199336413 63}{25000000000000000000000000000000} \leq p_{01}^R, \tag{34}$$

$$p_{02}^L \leq -\frac{2493692896887403399408861867 1}{10^{30}} < 0. \tag{35}$$

Thus the bridge sections do not vanish. The overlap intervals

$$\left[\frac{79157}{10000}, \frac{31663}{4000}\right], \qquad \left[\frac{158317}{20000}, \frac{79159}{10000}\right]$$

are nonempty, and the transition function is the ratio of two sections already proved nonzero there. Dividing by the chosen section normalizes it to one. This proves that the atlas has no uncovered endpoint or transition point.

The repeated-multiplier energy y_* lies in the right outer chart. The confluent formula for Ω_{013}, together with the nonvanishing of the selected section there, proves that the physical plane and its matching determinant remain represented at y_*. Equivalently, the root of $12y^2 - 96y - 17$ in I is a regular point of the Grassmann curve even though the individual multiplier labels meet.

A.5 Exhaustion and genuine unsquared exclusion

Every zero of the physical determinant on I must annihilate the resultant (31). The Sturm table above shows that only p_6 and q_6 have roots in I, one each. They are isolated by

$$I_1 = \left[\frac{8080985802104273}{10^{15}}, \frac{4040492901052137}{5 \cdot 10^{14}}\right], \tag{36}$$

$$I_2 = \left[\frac{406992828166963}{5 \cdot 10^{13}}, \frac{101748207041741}{12500000000000}\right]. \tag{37}$$

There are no other resultant roots on I. The chart-section root near 7.915780041490243 is not in this list because the bridge chart shows it to be a coordinate degeneracy only. The repeated-multiplier value (21) is represented by the confluent quotient and is not a missing resultant root.

Here is the cross-chart necessity argument explicitly. At every $y \in I$, the atlas provides a chart in which both selected Plücker sections are nonzero. In that chart, the normalized numerator differs from the coordinate-free exterior determinant only by the product of those nonzero sections and the nonzero ordered Vandermonde factors. Hence a physical zero makes the chart numerator vanish. Clearing only the denominators already proved nonzero gives

$$R(P, y) = G(P, y) = 0,$$

so the resultant must vanish. At the sole section zero the middle chart supplies this implication; at y_* the confluent wedge replaces the Vandermonde quotient and gives the same implication. Thus neither a chart transition nor the repeated multiplier can evade the candidate list.

It remains to test the original unsquared equation, not its squared elimination. In chart 013, direct substitution of the physical branch (33) into (24) gives

$$\begin{aligned} -\frac{836276240139532242462895459561543314 1079}{5 \cdot 10^{29}} &\le \mathcal{E}_6(y), \\ \mathcal{E}_6(y) \le -\frac{836275924043504022741774193992955683 0821}{5 \cdot 10^{29}} &< 0 \quad (y \in I_1), \end{aligned} \tag{38}$$

$$\begin{aligned} -\frac{964632265935538380053083716179130270 3671}{5 \cdot 10^{29}} &\le \mathcal{E}_6(y), \\ \mathcal{E}_6(y) \le -\frac{192923981941593176156733543362325386 13017}{10^{30}} &< 0 \quad (y \in I_2). \end{aligned} \tag{39}$$

Both intervals are strictly negative. Reconstructing the same planes from cofactor rows 023 gives strictly negative rational enclosures on I_1 and I_2 as well. The latter reconstruction uses different Plücker sections and therefore also checks that the exclusion is not an artifact of the 013 normalization.

The root in I_1 is a physical root for a different defect transfer, but equation (38) shows that it is not a G6 match. The root in I_2 comes from the second algebraic branch retained by squaring and clearing denominators; equation (39) excludes it as well. Consequently the positive G6 determinant has no zero on I. Together with the unique physical root in J_6, the essential-spectrum reduction, and $\|A_6\| \le 4$, this proves that $+\sqrt{c_6}$ is the largest positive spectral point of A_6.

A.6 The symmetry and the squared multiplicity

We finish by auditing the negative branch and the multiplicity after squaring. Let

$$D_6^+ = \{-4j : j \ge 0\} \cup \{6 + 4j : j \ge 0\}$$

be the set of positive Q-sites. The reflection $i \mapsto 6 - i$ exchanges its two displayed parts, so

$$Q_{6-i} = Q_i \qquad (i \in \mathbb{Z}). \tag{40}$$

The finite anchor computation

$$(Q_0, \ldots, Q_6) = (1, -1, -1, -1, -1, -1, 1), \qquad (\tau_0, \ldots, \tau_7) = (1, 1, -1, 1, -1, 1, -1, -1)$$

gives $\tau_7 = -\tau_0$. If $\tau_{7-i} = -\tau_i$, then

$$\tau_{6-i} = Q_{6-i}\tau_{7-i} = -Q_i\tau_i = -\tau_{i+1}.$$

The same recurrence run backwards proves

$$\tau_{7-i} = -\tau_i \qquad (i \in \mathbb{Z}). \tag{41}$$

Define the unitary operator

$$(Ku)_i = (-1)^i u_{9-i}.$$

Then

$$(K^2u)_i = (-1)^i(-1)^{9-i}u_i = -u_i,$$

so $K^2 = -I$. Moreover,

$$\begin{aligned}(KA_6u)_i &= (-1)^i\big(u_{8-i} + u_{10-i} + \tau_{7-i}u_{7-i} + \tau_{9-i}u_{11-i}\big),\\ (A_6Ku)_i &= (-1)^i\big(-u_{10-i} - u_{8-i} + \tau_{i-2}u_{11-i} + \tau_i u_{7-i}\big).\end{aligned}$$

Using (41) at i and $i-2$ gives

$$KA_6 = -A_6K, \qquad KH_6 = H_6K.$$

Thus K maps the one-dimensional eigenspace at $+\sqrt{c_6}$ bijectively onto a one-dimensional eigenspace at $-\sqrt{c_6}$. It also maps any hypothetical negative spectral point of larger absolute value to a positive point already excluded above.

Finally, because A_6 is self-adjoint and $c_6 > 0$,

$$\ker(H_6 - c_6) = \ker(A_6 - \sqrt{c_6}) \overset{\perp}{\oplus} \ker(A_6 + \sqrt{c_6}). \tag{42}$$

Both summands have dimension one. Therefore

$$\sup\sigma(H_6) = c_6, \qquad \dim\ker(H_6 - c_6) = 2.$$

Reflection reverses the interface by unitary conjugacy, and negating the lift gives a diagonally conjugate squared operator. Hence the same global edge and rank-two conclusion hold for both orientations and both lifts.

B Exact Finite Classification and Completeness

This appendix gives the finite part of the classification as a mathematical proof. Its computational input consists only of finite integer matrices, integer vectors, rational inequalities, exact polynomial factorization, and rational positive-definiteness certificates. In particular, no conclusion below is inferred from a floating-point eigenvalue.

B.1 The finite quotient and its spectral certificates

We first make precise why the finite domains used below cover every signing. In the notation of Section 2, switching along a Hamilton path gives

$$a_0 = \cdots = a_{n-2} = 1, \qquad a_{n-1} = \alpha \in \{\pm 1\}.$$

The switching class is then determined by the triangle-flux word $\tau \in \{\pm 1\}^n$ and the holonomy α. On passing to

$$Q_i = \tau_i\tau_{i+1},$$

one obtains the necessary cyclic condition

$$\prod_{i=0}^{n-1} Q_i = 1. \tag{43}$$

Conversely, every cyclic Q-word satisfying (43), together with a choice of τ_0, reconstructs a unique cyclic lift through $\tau_{i+1} = Q_i\tau_i$, and either value of α then reconstructs a signing. Thus (Q, τ_0, α) covers all switching classes.

For even n, the two choices of τ_0 need not have the same unsquared spectrum, but equation (2) gives

$$A_{-\tau,\alpha} = -DA_{\tau,\alpha}D, \qquad A^2_{-\tau,\alpha} = DA^2_{\tau,\alpha}D, \qquad D = \operatorname{diag}\big((-1)^i\big)_{i=0}^{n-1}.$$

They therefore have identical squared spectra and spectral radii. It is legitimate for the present minimization problem to fix $\tau_0 = 1$, but it is not legitimate to omit either holonomy. We consequently use (Q, α) as a quotient only after this squared-spectral identification.

Rotation of Q is induced by relabelling the vertices. Reflection is also induced by a graph isomorphism; if the normalization $\tau_0 = 1$ is restored afterward, it may additionally negate the lift. Hence both operations preserve the squared spectrum and spectral radius. Passing to a dihedral representative loses no possible value of $\rho(A)$. Notice, however, that neither operation identifies the two values of α, which are retained separately throughout.

We use two elementary exact certificate principles. Since A is real symmetric,

$$\rho(A)^2 = \lambda_{\max}(A^2) = \max_{v\neq 0} \frac{\|Av\|^2}{\|v\|^2}. \tag{44}$$

Thus an integer vector $v \neq 0$ for which

$$\|Av\|^2 \geq \theta_n \|v\|^2 \tag{45}$$

proves $\rho(A)^2 \geq \theta_n$. The comparison in (45) is exact: θ_n is selected as an isolated real root of its integer minimal polynomial, so the sign of the algebraic difference is decided by rational interval isolation.

In the opposite direction, if positive integers p, q satisfy

$$pI - qA^2 \succ 0, \tag{46}$$

then every eigenvalue λ of A satisfies $p - q\lambda^2 > 0$, and hence

$$\rho(A)^2 < \frac{p}{q}. \tag{47}$$

Positive definiteness in (46) is decided over $\mathbb{Q}$, either from strictly positive fraction-free leading principal minors or from an exact factorization LDL^T with every diagonal entry of D positive.

Finally, for every even n,

$$\theta_n = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n} > 8 - \frac{20\pi^2}{n^2} > 8 - \frac{200}{n^2}. \tag{48}$$

The first strict inequality follows from $\cos x > 1 - x^2/2$ for nonzero x, and the second from $\pi^2 < 10$.

B.2 Complete decisions through order thirty

For $8 \leq n \leq 20$, direct enumeration uses one representative of each switching class. Since G_n is connected and has $2n$ edges, the number of switching classes is

$$\frac{2^{2n}}{2^{n-1}} = 2^{n+1}.$$

Table 1: Exact exhaustive decisions for even orders at most thirty. In the last five rows, a quotient record can represent an entire dihedral orbit.

n	Switching classes	Nonthreshold records	Threshold closure
8	512	510	2 exact equalities
10	2,048	2,046	2 exact equalities
12	8,192	8,190	2 exact equalities
14	32,768	32,766	2 exact equalities
16	131,072	131,070	2 exact equalities
18	524,288	524,286	2 exact equalities
20	2,097,152	2,097,150	2 exact equalities
22	8,388,608	97,467	threshold quotient
24	33,554,432	353,811	threshold quotient
26	134,217,728	1,299,063	threshold quotient
28	536,870,912	4,810,471	threshold quotient
30	2,147,483,648	17,929,599	threshold quotient

For $n = 22$, the same space is represented in the (Q, α) quotient described above. At $n = 24, 26, 28, 30$, cyclic Q-words are generated by weight and then reduced under rotations and reflections. Summing the represented dihedral-orbit sizes and retaining both values of α gives 2^n reduced (Q, α) states. Each such state represents the two choices $\tau_0 = \pm 1$, so the weighted switching-class total is 2^{n+1}. Their spectral decision may be shared only because the two lifts have identical squared spectra. Thus these are reorganizations of the full finite domain, not restricted searches.

Every represented class is discharged in one of two ways. A nonthreshold class carries a primitive integer vector satisfying (45). In a threshold class, the exact characteristic polynomial of A^2 contains the minimal polynomial of θ_n, and exact isolation of all real roots proves that this factor supplies the largest squared eigenvalue. Hence such a class has $\rho(A)^2 = \theta_n$. The complete finite decisions are summarized in Table 1.

It follows from the exhaustive alternative above that every signing at these orders satisfies $\rho(A)^2 \geq \theta_n$. Proposition 2.2 supplies, at every even order, a signing with $\rho(A)^2 = \theta_n$. Therefore

$$m_n = \rho_-(n) \qquad (8 \leq n \leq 30,\ n \text{ even}). \tag{49}$$

B.3 The exact counterexamples at orders thirty-two and forty

At order 32, take $\alpha = +1$ and the triangle-flux word

$$\tau = (+, +, -, +, -, -, +, -)^4.$$

The Hamilton-gauge reconstruction gives an integer symmetric signed adjacency matrix A_{32}. Exact fraction-free elimination, followed by an independent rational LDL^{T} factorization of the same reconstructed matrix, gives

$$1561 I_{32} - 200 A_{32}^2 \succ 0. \tag{50}$$

By (47), $\rho(A_{32})^2 < 1561/200$. Exact root isolation for θ_{32} gives

$$\frac{1561}{200} < \frac{11896117236720419}{1523321182060814} < \theta_{32}.$$

Consequently $m_{32} < \rho_-(32)$. The matrix inequality (50) is an upper certificate for this one explicit signing; it neither computes the exact value of m_{32} nor asserts that the displayed signing is the unique minimizer.

At order 40, encode $Q_i = +1$ by the digit 1 and take

$$Q = 1000100010001000100010001000100010001000, \qquad \alpha = -1,$$

with $\tau_0 = 1$. The recurrence $\tau_{i+1} = Q_i\tau_i$ and Hamilton gauge reconstruct a unique integer symmetric matrix A_{40}. Exact rational elimination gives forty positive pivots in

$$15541I_{40} - 2000A_{40}^2, \tag{51}$$

so

$$\rho(A_{40})^2 < \frac{15541}{2000}.$$

Moreover,

$$\frac{15541}{2000} < \frac{63}{8} = 8 - \frac{200}{40^2} < \theta_{40},$$

where the first difference is $209/2000$ and the last inequality is (48). Hence $m_{40} < \rho_-(40)$. This full-matrix upper certificate is logically independent of the lower-bound classification at the six intervening orders.

B.4 Local exclusion at the six recovery orders

We now prove the finite-state reduction used for

$$\mathcal{N} = \{34, 36, 38, 42, 44, 46\}.$$

For each $n \in \mathcal{N}$, exact cyclotomic elimination gives the minimal polynomial of θ_n. The value θ_n is its rightmost real root: if c is a conjugate of $\cos(2\pi/n)$, then the corresponding conjugate is

$$f(c) = 2 + 2c + 4c^2.$$

The function f is increasing on $[-1/4, 1]$, while $f(c) \le 4$ on $[-1, -1/4]$; the smallest nonzero angular distance among the cyclotomic conjugates is $2\pi/n$. Exact Sturm counts therefore select strict rational endpoints

$$a_n < \theta_n < b_n. \tag{52}$$

Let $S = \{0, \ldots, L-1\}$, where

$$L_{34} = 12, \qquad L_{36} = 13, \qquad L_{38} = L_{42} = L_{44} = L_{46} = 14. \tag{53}$$

Because these supports and their range-two collars are proper arcs of the corresponding cycles, the holonomy seam can be moved outside the collar by switching. For a vector supported on S, equation (1) defines an integer map

$$C_W : \mathbb{R}^L \longrightarrow \mathbb{R}^{L+4},$$

where the length-$(L+1)$ word

$$W = (Q_{-2}, Q_{-1}, \ldots, Q_{L-2})$$

determines all required τ-coefficients up to simultaneous negation. That negation does not change the squared spectrum. If P_S denotes compression to S, then

$$M_W := C_W^{\mathsf{T}} C_W = P_S A^2 P_S. \tag{54}$$

Lemma B.1 (Local exclusion)**.** *If an integer vector $v \neq 0$ satisfies*

$$v^{\mathsf{T}} M_W v > b_n v^{\mathsf{T}} v,$$

then every cyclic signing containing W satisfies $\rho(A)^2 > \theta_n$.

Proof. Extend v by zero off S. Equations (44) and (54) give

$$\rho(A)^2 \geq \frac{v^{\mathsf{T}} M_W v}{v^{\mathsf{T}} v} > b_n > \theta_n.$$

The argument is unaffected by the location of the window or the holonomy seam. □

Every one of the 2^{L+1} windows is decided by exact arithmetic. A surviving window satisfies

$$a_n I - M_W \succ 0,$$

as proved by the fraction-free Sylvester criterion; an excluded window carries an integer vector satisfying Lemma B.1. The two lists form a partition of all windows. Thus a signing is either already excluded by a local Rayleigh certificate or every one of its cyclic windows belongs to the finite surviving set.

B.5 Parity-lifted overlap closure

Encode $Q_i = +1$ by $b_i = 1$ and $Q_i = -1$ by $b_i = 0$. Since n is even, condition (43) becomes

$$\sum_{i=0}^{n-1} b_i \equiv 0 \pmod 2. \tag{55}$$

Let $\mathcal{W}_n$ be the surviving binary words of length $L+1$. Construct a directed overlap graph with length-L states: a word $(b_0, \ldots, b_L) \in \mathcal{W}_n$ is the edge

$$(b_0, \ldots, b_{L-1}) \longrightarrow (b_1, \ldots, b_L).$$

Lift every state to (s, ε), with $\varepsilon \in \mathbb{Z}/2\mathbb{Z}$, and lift the edge by

$$(s, \varepsilon) \longrightarrow (s', \varepsilon + b_L). \tag{56}$$

Lemma B.2 (Soundness and completeness)**.** *Length-n closed walks in the parity-lifted overlap graph are in bijection with rooted cyclic Q-words satisfying* (43) *and containing no excluded window.*

Proof. Along a walk, consecutive edges overlap in exactly L symbols, so their appended labels form a word of length n. Closure of the base state forces the last L overlaps with the first L; hence every wraparound window also survives. Closure of the parity coordinate in (56) says that the sum of all appended bits is even, which is precisely (55). This constructs a legal cyclic Q-word and proves soundness.

Conversely, start at any root of a legal cyclic Q-word having no excluded window and read its symbols consecutively. Each length-$(L+1)$ window is an edge of the overlap graph, successive windows have the required overlap, cyclicity closes the base state, and (55) closes the parity state. This produces a length-n closed walk and proves completeness. □

Table 2: Exact finite-state closure at the six recovery orders.

n	L	Surviving	States	Rooted even	Rooted odd	Canonical Q	Terminals
34	12	124	92	1	0	1	2
36	13	128	92	1	4	1	2
38	14	184	132	77	38	3	6
42	14	232	166	337	392	7	14
44	14	240	171	353	620	10	20
46	14	240	171	599	690	10	20

After enumerating these closed walks, rotation and reflection identify different roots and orientations of the same cyclic configuration. By the spectral equivalences proved in Subsection B.1, taking one dihedral representative is exhaustive. For every resulting Q-word, both $\alpha = -1$ and $\alpha = +1$ are retained. The two τ-lifts are already covered by their identical squared spectra. Consequently the terminal (Q, α)-records cover every signing that was not locally excluded.

The exact finite counts are given in Table 2. The rooted even and odd columns count closed walks by total bit parity before the cyclic and dihedral reductions; only the even walks are globally liftable. Different roots can represent the same cyclic word.

There are therefore exactly

$$2(1 + 1 + 3 + 7 + 10 + 10) = 64 \tag{57}$$

terminal (Q, α)-records. Six of them have the all-negative Q-word and $\alpha = -1$. For each of these, the minimal polynomial of θ_n divides $\det(xI - A^2)$, and exact root isolation shows that this factor gives the largest squared eigenvalue. Thus these six terminals satisfy $\rho(A)^2 = \theta_n$. Each of the other 58 terminals carries a primitive integer vector v with

$$\frac{\|Av\|^2}{\|v\|^2} > b_n > \theta_n.$$

All 64 terminals are therefore decided, and none remains unresolved.

Every signing at one of the six recovery orders either contains an excluded window, in which case Lemma B.1 applies, or gives a parity-even closed walk, in which case Lemma B.2 places it in one of the resolved terminals. Hence

$$\rho(A)^2 \geq \theta_n \qquad (n = 34, 36, 38, 42, 44, 46).$$

Combining this universal lower bound with Proposition 2.2 yields

$$m_n = \rho_-(n) \qquad (n = 34, 36, 38, 42, 44, 46). \tag{58}$$

B.6 The exact bridge from forty-eight to two hundred thirty-eight

It remains to record precisely the finite boundary between the isolated small orders and the analytic tail. The even interval

$$48 \leq n < 240$$

contains exactly $(238-48)/2+1=96$ orders. At each such order, define A_n by the deterministic residue family

$n \bmod 8$	flux construction	α
0	period-eight repetition	-1
2	one G6 phase slip	$+1$
4	two balanced G6 phase slips	-1
6	three balanced G6 phase slips	$+1$.

The gap words are those of Section 6; the cyclic lift condition is checked before Hamilton gauge, so every row determines an actual signed cycle square rather than an open-chain approximation.

For every even n in this interval, the finite certificate supplies positive integers p_n, q_n such that

$$p_n I - q_n A_n^2 \succ 0, \qquad \frac{p_n}{q_n} < 8 - \frac{200}{n^2}. \tag{59}$$

Each matrix A_n is reconstructed from its displayed residue rule, and each first inequality is verified by an exact rational LDL^{T} factorization with positive diagonal. Thus the finite statement consists of all 96 consecutive even orders and has no interpolation or sampling step. Equations (47), (59), and (48) give, separately for every row,

$$\rho(A_n)^2 < \frac{p_n}{q_n} < 8 - \frac{200}{n^2} < \theta_n.$$

Therefore

$$m_n < \rho_-(n) \qquad (48 \le n < 240,\ n \text{ even}). \tag{60}$$

The endpoint 238 is included in this exact finite bridge. The next even order, 240, is the first order handled by the residue-wise IMS propagation in Section 6; no order lies between the two arguments.

Equations (49), (58), the strict witnesses at 32 and 40, and (60) are exactly the finite assertions used in the proof of Theorem 1.1.

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